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# MD. Jakirul Islam

Aspiring AI & Software Developer | CSE Undergraduate

## PROFESSIONAL SUMMARY

Motivated 5th-semester Computer Science & Engineering undergraduate at Daffodil International University with strong foundations in C, C++, Data Structures, Algorithms, and Object-Oriented Programming. Passionate about architecting efficient low-level systems, tackling algorithmic challenges in competitive programming, and actively expanding into AI/Machine Learning solutions.

## EDUCATION

|                                                     |                      |
|-----------------------------------------------------|----------------------|
| <b>B.Sc. in Computer Science &amp; Engineering</b>  | Expected Fall 2028   |
| Daffodil International University (DIU), Dhaka      | 5th Semester Student |
| <b>Higher Secondary Certificate (HSC) - Science</b> | 2022                 |
| National College of Education                       | GPA 5.00 / 5.00      |
| <b>Secondary School Certificate (SSC) - Science</b> | 2020                 |
| Narsingdi Model School                              | GPA 4.94 / 5.00      |

## TECHNICAL PROJECTS

### Hall-Canteen-Portal C CLI Structs File I/O

- Designed and developed a terminal-based university hall management ecosystem featuring role-based authentication for students, staff, and administrators.
- Implemented real-time room allocation matrix, automated token generation for canteen dining, and dynamic monthly billing persistence using custom C structures and binary files.

### Hall Student Management System C Linked Lists Dynamic Memory

- Engineered a high-performance student record management system utilizing Singly Linked Lists for dynamic node allocation, insertions, deletions, and search operations.
- Eliminated fixed memory constraints by leveraging heap allocation (malloc/free) and integrated persistent file I/O for saving and restoring records across sessions.

### Hospital Queue Manager C Priority Queue Greedy Algorithm ANSI UI

- Built an emergency medical triage priority management system applying greedy heuristics to sort and dispatch patients based on clinical criticality levels.
- Crafted an interactive ANSI color-coded console interface allowing healthcare staff to monitor real-time queue states and patient vitals.

## ACHIEVEMENTS & EXTRACURRICULARS

- Finalist, Take-off Programming Contest** — Competed and solved complex algorithmic problems under tight time bounds.
- Contestant, Unlock the Algorithm** — Participated in university-wide competitive programming contest demonstrating proficiency in graph and array manipulation.
- Math Olympiad Participant & Awardee** — Demonstrated strong analytical thinking and discrete mathematical problem-solving skills.

## TECHNICAL SKILLS

Languages: C, C++, Java, Python

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|--------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Core Concepts:</b>    | Data Structures, Algorithms, Object-Oriented Programming (OOP), Dynamic Memory Management, AI/ML (Foundations) |
| <b>Tools &amp; IDEs:</b> | VS Code, Codeblocks, Git, GitHub, Terminal / Bash CLI, Linux                                                   |